

⚡ Episteme — STEM Knowledge Assistant

An intelligent AI-powered STEM knowledge assistant built with RAG (Retrieval-Augmented Generation), featuring multi-modal learning tools and Google OAuth authentication.

Show Image

Show Image

Show Image

Show Image

🚀 Features

Feature	Description
🧠 RAG Pipeline	Retrieves from Wikipedia, ArXiv, PubMed + uploaded PDFs
🔒 Google OAuth	Secure login — each user has private chat history
📊 Diagram Generator	Auto-generates Mermaid.js diagrams
🖋️ Quiz Mode	MCQ quiz with scoring and explanations
📏 Compare Mode	Side-by-side comparison tables
🧑 ELI5 Mode	Explains complex topics simply
💻 Code Explainer	Analyzes and explains code
📐 Formula Renderer	LaTeX math formula rendering
🔍 ArXiv Search	Finds verified research paper links
📄 Session Summary	Auto-generates PDF session summaries
🟢 Confidence Meter	Shows answer confidence level
💡 Follow-up Questions	Suggests relevant next questions
📄 PDF Upload	Ask questions from uploaded documents
🌙 Dark/Light Theme	Toggle between themes

Tech Stack

Frontend	→ Streamlit
LLM	→ Groq (LLaMA 3.3 70B)
Embeddings	→ HuggingFace (all-MiniLM-L6-v2)
Vector DB	→ ChromaDB
Framework	→ LangChain
Auth	→ Google OAuth 2.0
Search	→ ArXiv API + DuckDuckGo
PDF Reports	→ ReportLab
Database	→ SQLite

Project Structure

```
Episteme/  
├─ app.py           # Main Streamlit app  
├─ generator.py     # LLM generation + mode detection  
├─ retriever.py     # Hybrid RAG retrieval  
├─ ingest.py        # Knowledge base ingestion  
├─ pdf_loader.py    # PDF processing  
├─ database.py      # SQLite chat history  
├─ requirements.txt # Dependencies  
├─ .gitignore       # Hidden files  
└─ README.md        # This file
```

Setup & Installation

1. Clone the repository

```
bash  
  
git clone https://github.com/YOUR_USERNAME/Episteme.git  
cd Episteme
```

2. Create virtual environment

```
bash
```

```
python -m venv venv
venv\Scripts\activate # Windows
source venv/bin/activate # Mac/Linux
```

3. Install dependencies

```
bash

pip install -r requirements.txt
```

4. Set up environment variables

Create a `.env` file:

```
GROQ_API_KEY=your_groq_api_key
GOOGLE_CLIENT_ID=your_google_client_id
GOOGLE_CLIENT_SECRET=your_google_client_secret
SERPER_API_KEY=your_serper_api_key # optional
```

5. Set up Google OAuth

- Create project on [Google Cloud Console](#)
- Enable OAuth 2.0
- Download `client_secret.json` to project folder

6. Build knowledge base

```
bash

python ingest.py
```

7. Run the app

```
bash

streamlit run app.py
```

Special Modes

Command	Description
<code>Quiz me on PID controller</code>	Generates 5 MCQ questions
<code>Compare FPGA vs Arduino</code>	Side-by-side comparison table
<code>ELI5: Quantum computing</code>	Simple explanation with analogy
<code>Explain this code: [paste]</code>	Code analysis and bug detection
<code>Draw diagram of PID controller</code>	Auto-generates flowchart
<code>I need paper: YOLO</code>	Searches ArXiv for verified links

Privacy & Security

- All API keys stored in `.env` (never committed to GitHub)
- Google OAuth for secure authentication
- Each user has private, isolated chat history
- No data sharing between users

Developer

Ifrah Gohar Khan

-  BE Avionics Engineering — Institute of Space Technology, Islamabad
-  LinkedIn: [linkedin.com/in/ifrah-gohar](https://www.linkedin.com/in/ifrah-gohar)
-  GitHub: github.com/YOUR_USERNAME

License

This project is for educational and portfolio purposes.

